

Yufeng Wang

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parrwnago.github.io

Education

University of Pennsylvania

Master of Science in Engineering, Mechanical Engineering, GPA: 4.0/4.0

Philadelphia, PA
Aug 2025 – May 2027 (Expected)

University of Pittsburgh

Bachelor of Science in Engineering, Mechanical Engineering, GPA: 3.895/4.0

Pittsburgh, PA
Aug 2023 – May 2025

Sichuan University

Bachelor of Engineering, Mechanical Engineering, GPA: 3.56/4.0

Chengdu, China
Sep 2021 – Jul 2023

Research Experience

Spherical Support-Caster Integration for Off-Road Holonomic Robot

Aug 2025 – Present

Independent Study with Prof. Mark Yim, University of Pennsylvania

- Integrated passive split spherical support caster with independently rotating hemispheres and transverse rollers into HAMR/CHUTNI; designed and iterated mounting interfaces to meet caster height, chassis clearance, and terrain contact requirements.
- Debugged and validated ROS 2 closed-loop waypoint tracking on hardware using Vicon feedback; corrected base-frame and yaw alignment and tuned wheel and heading controllers for reliable trajectory execution.
- Built rosbag2 and Vicon evaluation pipeline to log robot pose, reference trajectories, tracking errors, and actuator commands and quantify longitudinal and lateral error, speed, yaw drift, path length, and trajectory RMSE.
- Developing a controlled comparison protocol for split spherical and conventional trailing casters across forward, reverse, lateral, and rotational motion, direction changes, and uneven terrain traversal.

Projects

Quadrotor Autonomy Stack with Local Replanning

Spring 2026

MEAM 620: Robotics, Course Project, University of Pennsylvania

- Developed full quadrotor autonomy stack integrating 3D occupancy grid A* planning, Ramer-Douglas-Peucker waypoint simplification, smooth polynomial trajectory generation, SE(3) control, and visual inertial state estimation.
- Extended planner with receding horizon local replanning using finite range occupancy map updates, guide path local goals, and fallback behavior for blocked or unreachable targets.
- Tuned map specific trajectory and controller parameters for reliable flight through maze, window, stairwell, slalom, over-under, and switchback environments.
- Achieved full autograder performance on all six main-task maps and completed extra-credit local replanning challenge.

Autonomous Competition Robot for Wall Following and Tower Interaction

Fall 2025

MEAM 5100: Design of Mechatronic Systems, Final Competition Project, University of Pennsylvania

- Led mechanical design and integration of autonomous competition robot, developing chassis layout, drivetrain packaging, ramp capable suspension and clearance features, and mounting structures for motors, electronics, sensors, and tower interaction mechanisms under competition size constraints.
- Debugged Arduino/C++ wall following controller on ESP32 using ToF and ultrasonic sensor feedback and differential motor correction for stable boundary tracking.
- Contributed to waypoint navigation and tower interaction logic using Vive tracking; coordinated sensor placement, mechanical clearance, and control behavior for ramp traversal, target alignment, and reliable competition task execution.

Autonomous Cube-Stacking Robot Arm

Fall 2025

MEAM 5200: Intro to Robotics, Final Project, University of Pennsylvania

- Built autonomous robot-arm pipeline to detect, grasp, and stack static and dynamic cubes in a timed competition task.

- Used camera-based block detection, FK/IK, tuned grasp poses, and gripper-width feedback to execute reliable pick-and-place stacking.
- Developed dynamic-block capture logic with safe transition poses, timed wait-and-grasp behavior, retry handling, and stack-height tuning.
- Implemented supporting collision-aware planners, including potential-field planning and RRT, for obstacle-aware joint-space motion.

Design of Desktop Tensile Test Machine

Mar 2024 – Apr 2024

Mechanical Design 2, Supervisor: Prof. Stephen Ludwick, University of Pittsburgh

- Oversaw gear design, stress analysis, and buckling analysis of the lead screw.
- Conducted literature review to collect different calculation methods for stress analysis.
- Built formulas in Excel to automatically calculate stress, torque, and rotational speed.
- Received the highest design score in the class.

Airsoft Gun Gearbox Analysis

Mar 2024 – Apr 2024

Mechanical Design 2, Supervisor: Prof. Stephen Ludwick, University of Pittsburgh

- Disassembled airsoft gun to identify operational functions.
- Created Excel table for stress analysis calculations and bill of materials.
- Wrote report on stress analysis of gear teeth and analyzed failure points.

Work Experience

Chenzhi Technology

Chongqing, China

Engineer Assistant, Steering System Design Department

May 2024 – Aug 2024

- Designed steering system housings in CATIA V5 according to manufacturer specific packaging and performance requirements.
- Assembled and debugged steering systems; adjusted rack and pinion clearance, assist motor position, and belt tension to improve mechanical fit and system response.
- Evaluated cast component integrity using X-ray inspection and compared roller bearings from multiple suppliers to assess system noise and vibration.
- Translated technical specifications for three overseas orders, including hardpoint geometry and temperature tolerances, and coordinated with clients to confirm interface data.

Leadership

English Association

Chengdu, China

Minister of Publicity Department, Sichuan University

Sep 2022 – Jun 2023

- Coordinated and executed four substantial student communication meetings.
- Held short lecture on writing skills for the TOEFL test.
- Conducted new member recruitment interviews and made selection decisions.
- Arranged publicity activities before each event, made posters, leaflets, and invitations, coordinated with the planning department, and updated social media after events.

Skills

Programming: Python, C++, MATLAB, Java

Robotics & Control: ROS 2, Gazebo, Vicon Motion Capture, rosbag2, PID Control, SE(3) Control, A*, RRT, VIO

Mechanical & Embedded: SolidWorks, CATIA V5, Arduino, ESP32, ToF and Ultrasonic Sensors